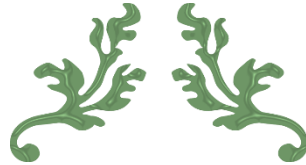


HOLY ANGEL UNIVERSITY
SCHOOL OF ENGINEERING AND ARCHITECTURE
DEPARTMENT OF CIVIL ENGINEERING



MODULE 1-PRELIM

Lesson 8: Influence Line-Beam



Structural Analysis

LESSON #8: Influence Line-Beam

- In this lesson you will be able to draw influence line for a statically determinate structure, also to determine the maximum live shear and moment in a member.

All concepts sample problems and figures are from the followings books,

- Hibbeler, R. C.(2017). Structural analysis 9th ed. Boston, Pearson
- Leet, Kenneth M. (2011). Fundamentals of structural analysis, McGraw-Hill, Boston
- Kassimali, Aslam (2010). Structural analysis, Cengage Learning, Stamford, CT USA
- J Rungamornrat (2011). Fundamental of Structural Analysis, Rungamornrat, Bangkok Thailand

You can also refer to the following online pages for additional information.

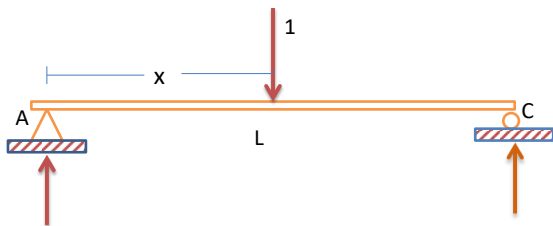
- HAU BOOKs 24x7
- <http://www.civilengineering.com>
- CE 474 Structural Analysis II Homepage.<https://engineering.purdue.edu/~ce474/>
- NPTEL Phase II::Civil Engineering-Structural Analysis I .nptel.ac.in/downloads/105101085/

Influence Line:

- Is a graph of response function of a structure as a function of the position of a downward unit load moving across the structure.
- It represents the variation of either the reaction, shear, moment, or deflection at a *specific point* in a member as a concentrated force moves over the member.
- Its best to illustrate the influence line through sample,

Steps: Influence line for the Reactions.

Step 1. Place a unit load (a load whose magnitude is equal to one) at a point, x, along the member.



Step 2. Use the equations of equilibrium to find the value of the reaction, shear, or moment at a specific point due the concentrated load as a function of x.

$$\sum M_C = 0$$

$$R_A(L) - 1(L - x) = 0$$

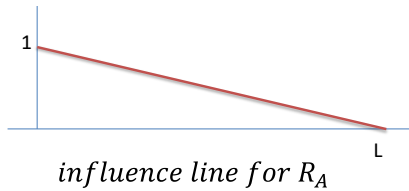
$$R_A = 1 - \frac{x}{L} \quad R_C = \frac{x}{L}$$

Step 3. Plot the values of the reaction, shear, or moment for the member.

For Reaction at A:

$$\text{When } x = 0, \quad R_A = 1$$

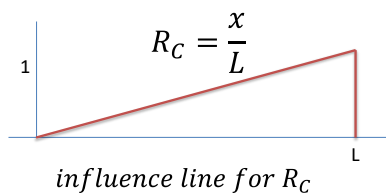
$$x = L, \quad R_A = 0$$



For Reaction at C.

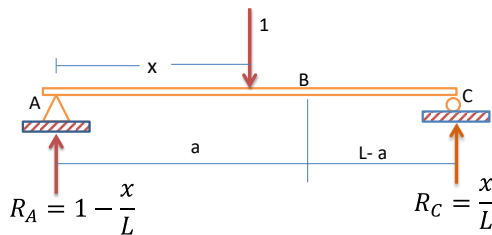
When $x = 0$, $R_C = 0$

When $x = L$, $R_C = 1$

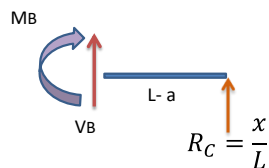


Influence line for Shear and Moment at B.

Step1.



Step 2. Consider the right side



$0 \leq x < a$

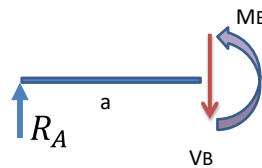
$$\sum F_y = 0$$

$$V_B = -R_C$$

$$+\sum M_B = 0$$

$$M_B = R_C(L - a)$$

Consider left side



$a < x \leq L$

$$\sum F_y = 0$$

$$V_B = R_A$$

$$+\sum M_B = 0$$

$$M_B = R_A(a)$$

Step 3.

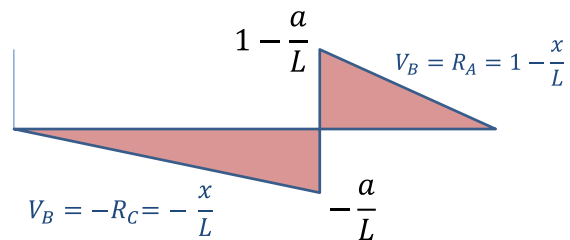
Influence Line for Shear at B.

$0 \leq x < a$

$$V_B = -R_C = -\frac{x}{L}$$

$a < x \leq L$

$$V_B = R_A = 1 - \frac{x}{L}$$



Influence Line for Moment at B

$0 \leq x < a$

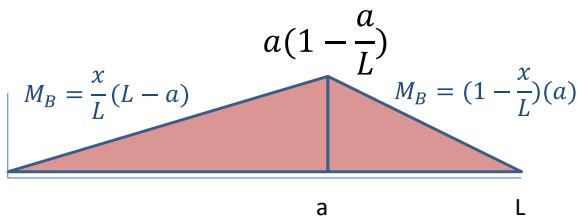
$$M_B = R_C(L - a)$$

$$M_B = \frac{x}{L}(L - a)$$

$a < x \leq L$

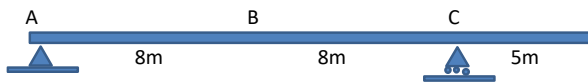
$$M_B = R_A(a)$$

$$M_B = (1 - \frac{x}{L})(a)$$



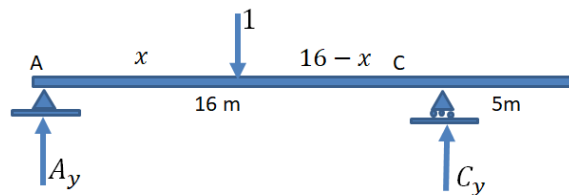
Problem #1:

Draw the influence lines for vertical reactions at supports A and C and the shear and bending moment at point B of the beams shown.



Solution

Influence line for reaction at A and C



$$0 \leq x \leq 16$$

$$\text{When } x = 0, \quad A_y = 1$$

$$+\uparrow \sum M_C = 0$$

$$C_y = 0$$

$$-1(16 - x) + A_y(16) = 0$$

$$\text{When } x = 16, \quad A_y = 0$$

$$A_y = \frac{16 - x}{16}$$

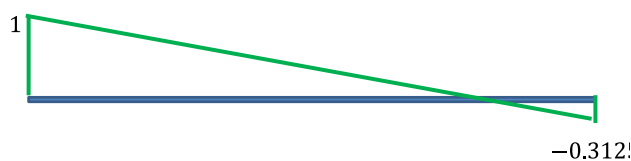
$$C_y = 1$$

$$C_y = \frac{x}{16}$$

$$\text{When } x = 21, \quad A_y = -0.3125$$

$$C_y = 1.3125$$

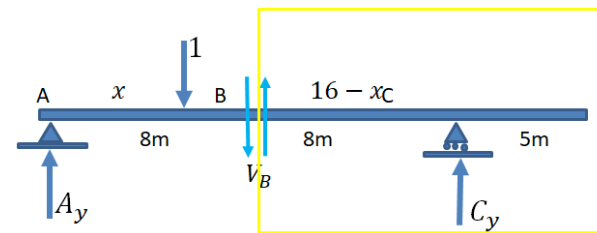
Reaction at A



Reaction at C



Influence line for Shear at B

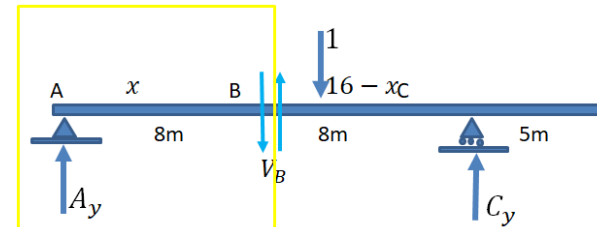


$0 \leq x \leq 8$, consider right side

$$+\uparrow \sum F_y = 0$$

$$V_B + C_y = 0$$

$$V_B = -C_y = -\frac{x}{16}$$



$8 \leq x \leq 21$, consider Left side

$$+\uparrow \sum F_y = 0$$

$$-V_B + A_y = 0$$

$$V_B = A_y = \frac{16 - x}{16}$$

$$0 \leq x \leq 8, \quad 8 \leq x \leq 21$$

$$V_B = -\frac{x}{16} \quad V_B = \frac{16 - x}{16}$$

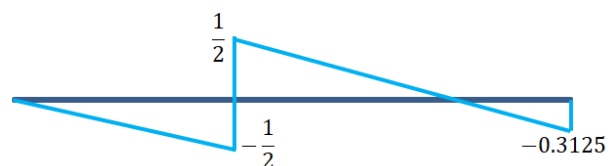
$$\text{when } x = 0, \quad x = 8$$

$$V_B = 0 \quad V_B = -\frac{1}{2}$$

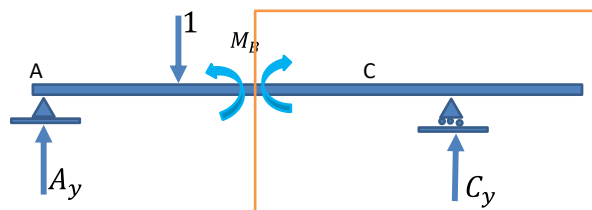
$$\text{when } x = 8, \quad x = 16$$

$$V_B = \frac{1}{2} \quad V_B = 0$$

$$\text{when } x = 21 \quad V_B = -0.3125$$



Influence line for moment at B



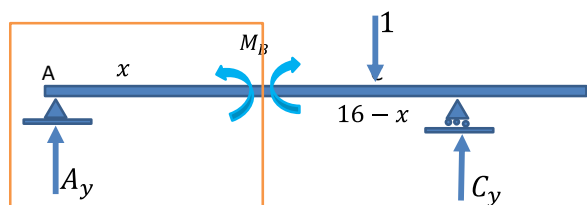
$0 \leq x \leq 8$, consider right side

$$\sum M_B = 0$$

$$M_B - C_y(8) = 0$$

$$M_B = C_y \left(\frac{x}{16} \right)$$

$$M_B = \frac{x}{2}$$



$8 \leq x \leq 21$, consider left side

$$\sum M_B = 0$$

$$-M_B + A_y(8) = 0$$

$$M_B = A_y \left(\frac{16-x}{16} \right)$$

$$M_B = \frac{16-x}{2}$$

$0 \leq x \leq 8$

when $x = 0$, $x = 8$

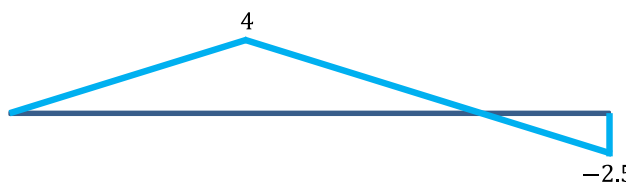
$$M_B = 0 \quad M_B = 4$$

$8 \leq x \leq 21$

when $x = 8$, $x = 16$

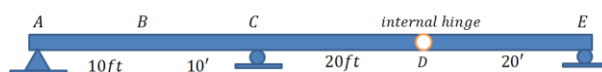
$$M_B = 4 \quad M_B = 0$$

when $x = 21$ $M_B = -2.5$



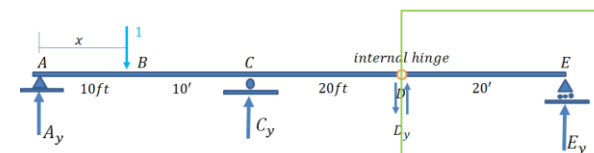
Problem#2

Draw the influence lines for the vertical reactions at supports A, C, and E, the shear just to the right of support C, and the bending moment at point B of the beam shown.



Solution:

influence line for Reactions A, C, E



When the 1 unit is at segment AC

for segment DE

$$D_y = E_y = 0$$

When the 1 unit is at segment AC

for segment ABCD

$$\sum M_A = 0$$

$$1(x) - C_y(20) + D_y(40) = 0$$

$$C_y = \frac{x}{20}$$

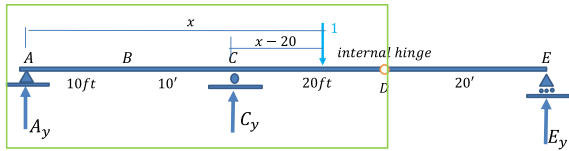
$$\sum M_C = 0$$

$$-1(20-x) + A_y(20) + D_y(20) = 0$$

$$A_y = \frac{20-x}{20}$$

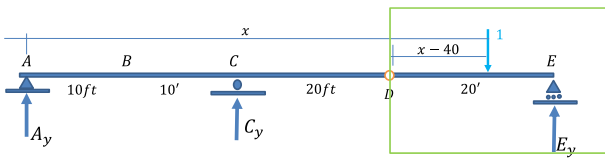
When the 1 unit is at segment CD

$$D_y = E_y = 0$$

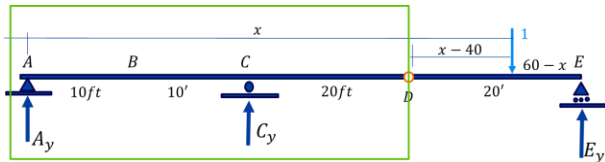


$$\begin{aligned} \sum M_A = 0 & \quad 0 & \sum M_C = 0 & \quad 0 \\ 1(x) - C_y(20) + D_y(40) = 0 & & 1(x - 20) + A_y(20) + D_y(20) = 0 \\ C_y = \frac{x}{20} & & A_y = \frac{20 - x}{20} \end{aligned}$$

When the 1 unit is at segment DE



$$\begin{aligned} \sum M_D = 0 & & \sum M_E = 0 \\ 1(x - 40) - E_y(20) = 0 & & -1(60 - x) + D_y(20) = 0 \\ E_y = \frac{x - 40}{20} & & D_y = \frac{60 - x}{20} \end{aligned}$$



$$\begin{aligned} \sum M_A = 0 & & C_y = \frac{60 - x}{20} (40) \left(\frac{1}{20} \right) \\ \frac{60 - x}{20} & & C_y = \frac{60 - x}{10} \\ D_y(40) - C_y(20) = 0 & & \end{aligned}$$

$$\begin{aligned} \sum M_C = 0 & \\ A_y(20) + D_y(20) = 0 & \\ A_y = -\frac{60 - x}{20} = \frac{x - 60}{20} & \end{aligned}$$

Influence Line for Reaction of A.

$$0 \leq x < 20$$

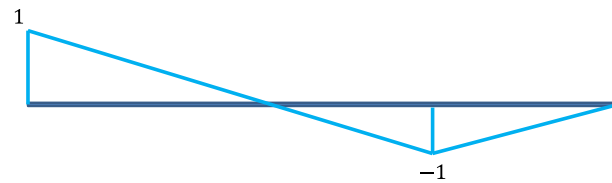
$$\begin{aligned} \text{when } x = 0, & & x = 20 \\ A_y = 1 & & A_y = 0 \end{aligned}$$

$$20 \leq x < 40$$

$$\begin{aligned} \text{when } x = 20, & & x = 40 \\ A_y = 0 & & A_y = -1 \end{aligned}$$

$$40 \leq x < 60$$

$$\begin{aligned} \text{when } x = 40, & & x = 60 \\ A_y = -1 & & A_y = 0 \end{aligned}$$



Influence Line for Reaction at C.

$$0 \leq x < 20$$

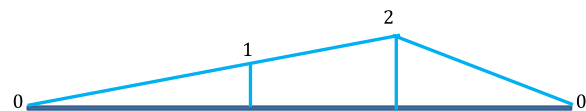
$$\begin{aligned} \text{when } x = 0, & & x = 20 \\ C_y = 0 & & C_y = 1 \end{aligned}$$

$$20 \leq x < 40$$

$$\begin{aligned} \text{when } x = 20, & & x = 40 \\ C_y = 1 & & C_y = 2 \end{aligned}$$

$$40 \leq x < 60$$

$$\begin{aligned} \text{when } x = 40, & & x = 60 \\ C_y = 2 & & C_y = 0 \end{aligned}$$



Influence Line for reaction at E.

Influence Line for reaction at E.

$$0 \leq x < 20$$

when $x = 0$, $x = 20$

$$E_y = 0 \quad E_y = 0$$

$$20 \leq x < 40$$

when $x = 20$, $x = 40$

$$E_y = 0 \quad E_y = 0$$

$$40 \leq x < 60$$

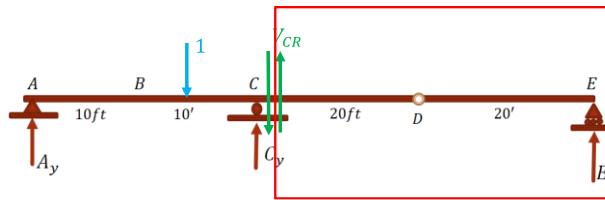
when $x = 40$, $x = 60$

$$E_y = 0 \quad E_y = 1$$



Influence line for Shear:

Shear just to the right of C

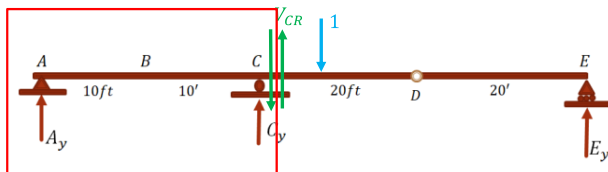


$$0 \leq x < 20, \text{ consider right side}$$

$$+\uparrow \sum F_y = 0$$

$$V_{CR} + E_y = 0$$

$$V_{CR} = 0$$

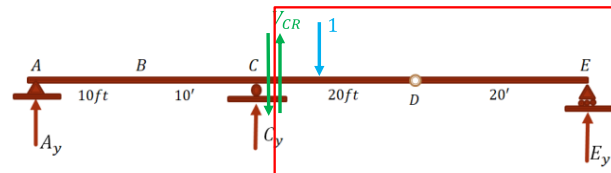


$$20 \leq x < 40, \text{ consider left side}$$

$$+\uparrow \sum F_y = 0 \quad A_y = \frac{20 - x}{20}$$

$$-V_{CR} + A_y + C_y = 0 \quad C_y = \frac{x}{20}$$

$$V_{CR} = 1$$



$$20 \leq x < 40, \text{ consider right side}$$

$$+\uparrow \sum F_y = 0$$

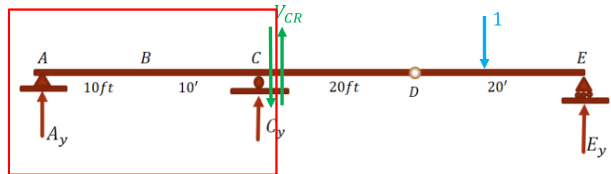
$$V_{CR} - 1 + E_y = 0$$

$$V_{CR} = 1$$

$$40 \leq x < 60, \text{ consider right side}$$

$$+\uparrow \sum F_y = 0 \quad V_{CR} = 1 - \frac{x - 40}{20}$$

$$V_{CR} - 1 + E_y = 0 \quad V_{CR} = \frac{60 - x}{20}$$



$$40 \leq x < 60, \text{ consider left side}$$

$$+\uparrow \sum F_y = 0 \quad V_{CR} = \frac{x - 60}{20} + \frac{60 - x}{10}$$

$$-V_{CR} + A_y + C_y = 0 \quad V_{CR} = \frac{60 - x}{20}$$

Shear just to the right of C

$$0 \leq x < 20 \quad V_{CR} = 0$$

when $x = 0$, $x = 20$

$$V_{CR} = 0 \quad V_{CR} = 0$$

Shear just to the right of C

$$20 \leq x < 40 \quad V_{CR} = 1$$

$$\text{when } x = 20, \quad x = 40$$

$$V_{CR} = 1 \quad V_{CR} = 1$$

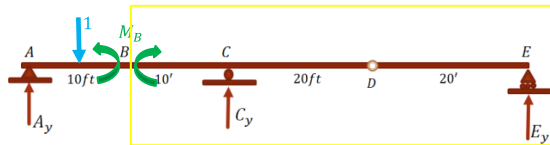
$$40 \leq x < 60 \quad V_{CR} = \frac{60 - x}{20}$$

$$\text{when } x = 40, \quad x = 60$$

$$V_{CR} = 1 \quad V_{CR} = 0$$



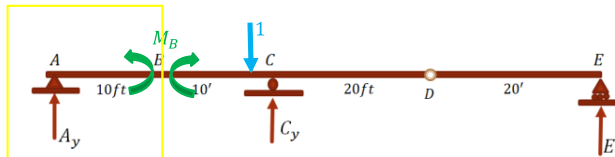
Bending Moment at B



$0 \leq x < 10$, consider right side

$$+\sum M_B = 0 \quad M_B = \frac{x}{20}(10)$$

$$M_B - C_y(10) - E_y(50) = 0 \quad M_B = \frac{x}{2}$$

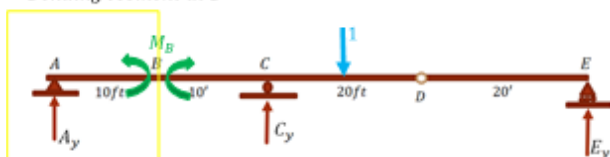


$10 \leq x < 20$, consider left side

$$+\sum M_B = 0 \quad M_B = \frac{20 - x}{20}(10)$$

$$-M_B + A_y(10) = 0 \quad M_B = \frac{20 - x}{2}$$

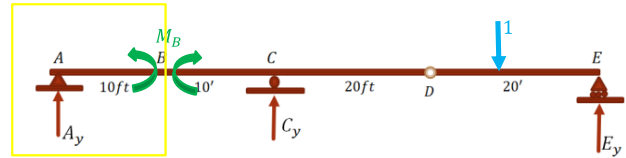
Bending Moment at B



$20 \leq x < 40$, consider left side

$$+\sum M_B = 0 \quad M_B = \frac{20 - x}{20}(10)$$

$$-M_B + A_y(10) = 0 \quad M_B = \frac{20 - x}{2}$$



$40 \leq x < 60$, consider left side

$$+\sum M_B = 0 \quad M_B = \frac{x - 60}{20}(10)$$

$$-M_B + A_y(10) = 0 \quad M_B = \frac{x - 60}{2}$$

$$0 \leq x < 10 \quad M_B = \frac{x}{2}$$

$$\text{when } x = 0, \quad M_B = 0$$

$$x = 10, \quad M_B = 5$$

$$10 \leq x < 20 \quad M_B = \frac{20 - x}{2}$$

$$\text{when } x = 10, \quad M_B = 5$$

$$x = 20, \quad M_B = 0$$

$$20 \leq x < 40 \quad M_B = \frac{20 - x}{2}$$

$$\text{when } x = 20, \quad M_B = 0$$

$$x = 40, \quad M_B = -10$$

$$40 \leq x < 60 \quad M_B = \frac{x - 60}{2}$$

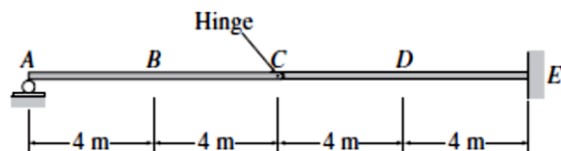
$$\text{when } x = 40, \quad M_B = -10$$

$$x = 60, \quad M_B = 0$$

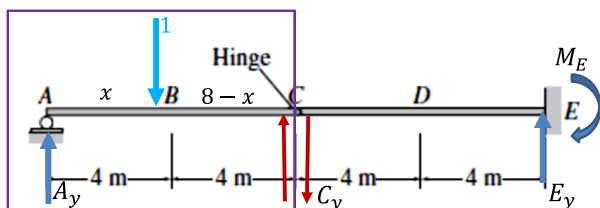


Problem#3: (P8.13-15, Structural Analysis, Aslam Kassimali, 4th ed.)

Draw the influence lines for the vertical reactions at supports A and E and the reaction moment at support E of the beam shown. Also the shear and bending moment at point B and D.

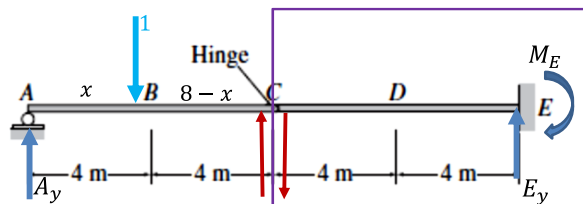


Solution



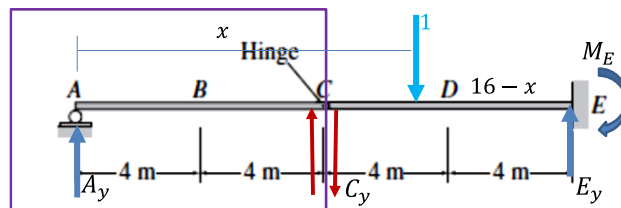
segment AC, $0 \leq x < 8$

$$\begin{aligned} \sum M_A = 0 \quad C_y = \frac{x}{8} \\ 1(x) - C_y(8) = 0 \end{aligned}$$



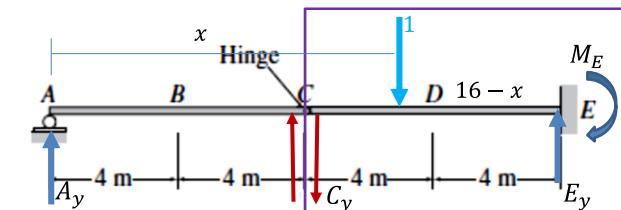
segment AC, $0 \leq x < 8$

$$\begin{aligned} \sum M_E = 0 \quad +\uparrow \sum F_y = 0 \\ M_E - C_y(8) = 0 \quad -C_y + E_y = 0 \\ M_E = x \quad E_y = \frac{x}{8} \end{aligned}$$



segment CE, $8 \leq x < 16$

$$A_y = C_y = 0$$



$$\begin{aligned} \sum M_E = 0 \quad +\uparrow \sum F_y = 0 \\ M_E - C_y(8) - 1(16 - x) = 0 \quad -C_y + E_y - 1 = 0 \\ M_E = 16 - x \quad E_y = 1 \end{aligned}$$

Influence Line for Reaction at A

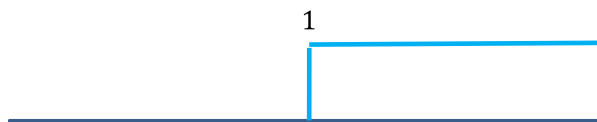
$$\begin{aligned} 0 \leq x < 8 \quad 8 \leq x < 16 \\ \text{when } x = 0, \quad A_y = 1 \quad \text{when } x = 8, \quad A_y = 0 \\ \text{when } x = 8, \quad A_y = 0 \quad \text{when } x = 16, \quad A_y = 0 \end{aligned}$$

Reaction at A



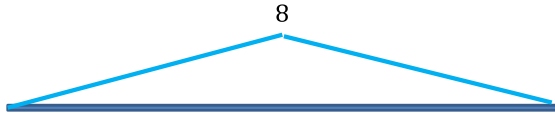
Influence Line for reaction at E.

$$\begin{aligned} 0 \leq x < 8 \quad 8 \leq x < 16 \quad E_y = 1 \\ \text{when } x = 0, \quad E_y = 0 \quad \text{when } x = 8, \quad E_y = 1 \\ \text{when } x = 8, \quad E_y = 1 \quad \text{when } x = 16, \quad E_y = 1 \end{aligned}$$

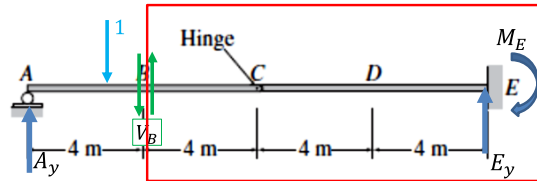


Influence Line for Moment at E

$$\begin{aligned}
 0 \leq x < 8 \quad M_E &= x & 8 \leq x < 16 \\
 \text{when } x = 0, \quad M_E &= 0 & \text{when } x = 8, \quad M_E &= 8 \\
 \text{when } x = 8, \quad M_E &= 8 & \text{when } x = 16, \quad M_E &= 0
 \end{aligned}$$



Influence Line for shear at B

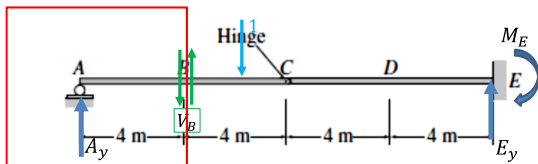


$0 \leq x < 4$, Consider right side

$$+\uparrow \sum F_y = 0 \quad E_y = \frac{x}{8}$$

$$V_B + E_y = 0$$

$$V_B = -E_y = -\frac{x}{8}$$

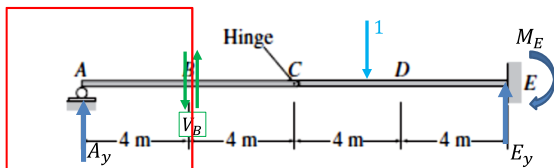


$4 \leq x < 8$, Consider left side

$$+\uparrow \sum F_y = 0 \quad A_y = \frac{8-x}{8}$$

$$-V_B + A_y = 0$$

$$V_B = A_y = \frac{8-x}{8}$$



$8 \leq x < 16$, Consider left side

$$+\uparrow \sum F_y = 0 \quad A_y = 0$$

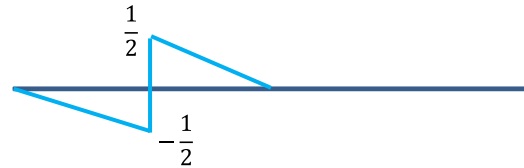
$$-V_B + A_y = 0$$

$$V_B = A_y = 0$$

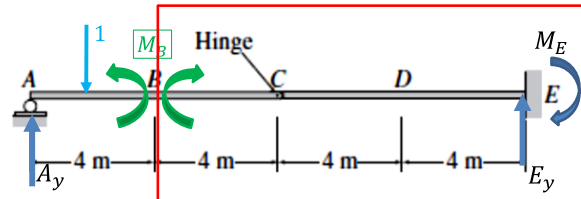
Influence Line for shear at B

$$\begin{aligned}
 0 \leq x < 4 \quad V_B &= -\frac{x}{8} & 4 \leq x < 8 \quad V_B &= \frac{8-x}{8} \\
 \text{when } x = 0, \quad V_B &= 0 & \text{when } x = 4, \quad V_B &= \frac{1}{2} \\
 \text{when } x = 4, \quad V_B &= -\frac{1}{2} & \text{when } x = 8, \quad V_B &= 0
 \end{aligned}$$

$$\begin{aligned}
 8 \leq x < 16 \quad V_B &= 0 \\
 \text{when } x = 8, \quad V_B &= 0 \\
 \text{when } x = 16, \quad V_B &= 0
 \end{aligned}$$



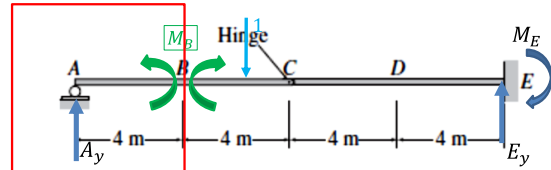
Influence Line for Moment at B



$0 \leq x < 4$, Consider right side

$$\begin{aligned}
 +\sum M_B &= 0 \quad \frac{x}{8} \\
 M_B + M_E - E_y(12) &= 0
 \end{aligned}$$

$$M_B = \frac{x}{2}$$

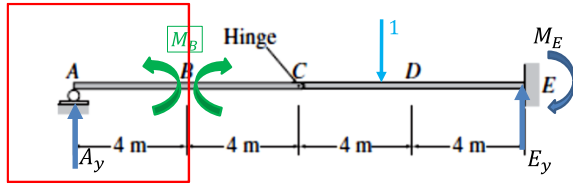


$4 \leq x < 8$, Consider left side

$$+\sum M_B = 0 \quad A_y = \frac{8-x}{8}$$

$$M_B = A_y(4)$$

$$M_B = \frac{8-x}{8}(4) = \frac{8-x}{2}$$



$8 \leq x < 16$, Consider left side

$$+\sum M_B = 0 \quad A_y = 0$$

$$M_B = A_y(4)$$

$$M_B = 0(4) = 0$$

$$0 \leq x < 4 \quad M_B = \frac{x}{2} \quad 4 \leq x < 8 \quad M_B = \frac{8-x}{2}$$

when $x = 0$, $M_B = 0$ when $x = 4$, $M_B = 2$

when $x = 4$, $M_B = 2$ when $x = 8$, $M_B = 0$

$$8 \leq x < 16 \quad M_B = 0$$

when $x = 8$, $M_B = 0$

when $x = 16$, $M_B = 0$



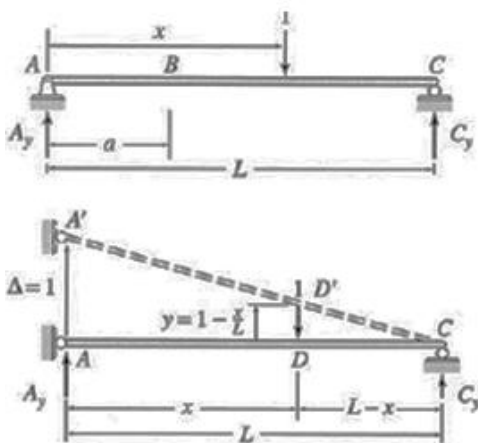
MULLER-BRESLAU'S PRINCIPLE AND QUALITATIVE INFLUENCE LINES

Heinrich Muller-Breslau In 1886

- The influence line for a force or moment response function is given by the deflected shape of the released structure obtained by removing the restraint corresponding to the response function from the original structure and by giving the released structure a unit displacement (or rotation) at the location and in the direction of the response function, so that only the response function and the unit load perform external work.
- This principle is valid only for the influence lines for response functions involving forces and moments (e.g., reactions, shears, bending moments, or forces in truss members), and it does not apply to the influence lines for deflections.

Say, Influence Line for Reaction at A.

Remove the ability to resist movement in the vertical direction at A by using a guided roller

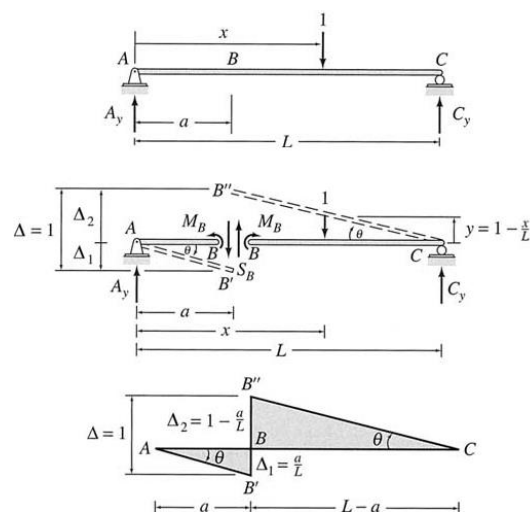


Influence Line of A_y

Fig. 8.8 a,b, Structural Analysis, Aslam Kassimali, 4th ed.

Influence Line for shear at point B.

Remove the ability to resist shear at point B.



(d) Influence Line for V_B

Fig. 8.8 a,d, Structural Analysis, Aslam Kassimali, 4th ed.

Influence Line for Moment at point B.

Remove the ability to resist moment at B by using a hinge.

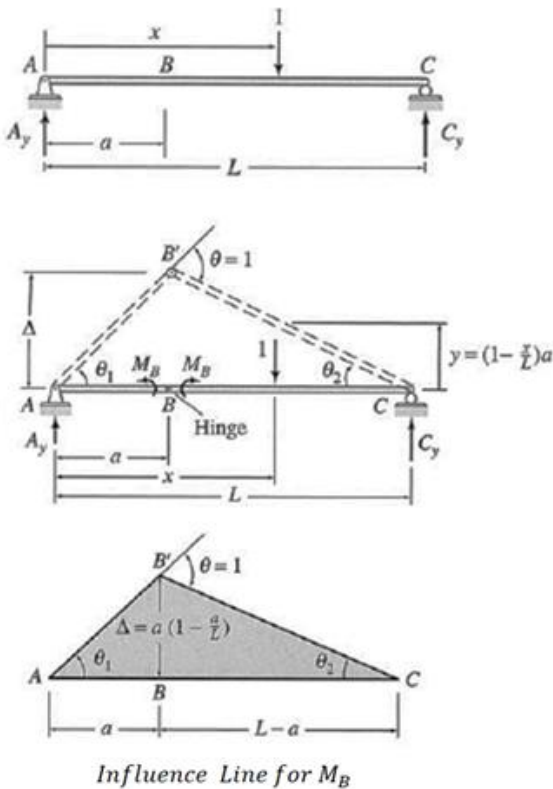
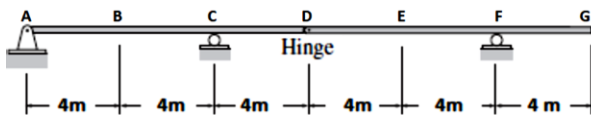


Fig. 8.8 a,e, Structural Analysis, Aslam Kassimali 4th ed.

Problem #3:

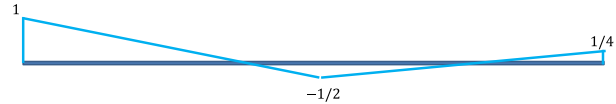
Draw the influence lines for the a) Vertical reaction at supports A, C, and F, b) Shear and bending moment at point B and, c) Shear and bending moment at point E of the beam shown.



Solution:

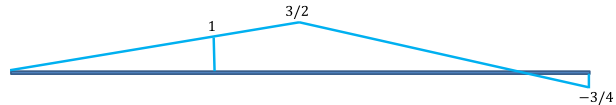
Influence Line for Reaction at A.

Release A and drag 1 unit upward, take note that there is no movement at the support.



Influence Line for Reaction at C.

Release C and drag 1 unit upward, take note that there is no movement at the support.

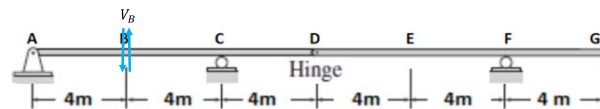


Influence Line for Reaction at E.

Release E and drag 1 unit upward, take note that there is no movement at the support.

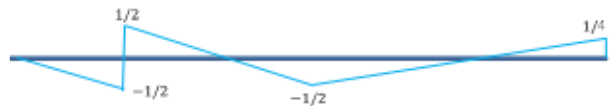


Influence Line for Shear at B.

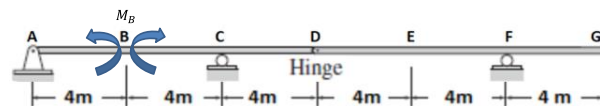


Remove the ability to resist shear at point B. Take note that there is no movement at the support.

Drag along the direction of force.

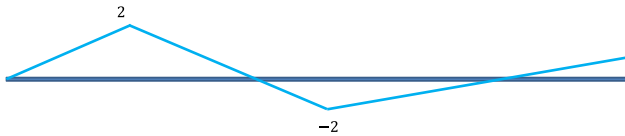


Influence Line for Moment at B

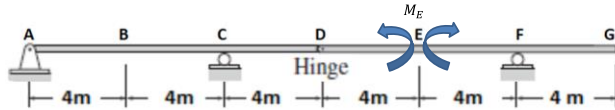


Remove the ability to resist Moment at point B. Take note that there is no movement at the support.

Rotate wrt the specified direction of moment.



Influence Line for Moment at E.

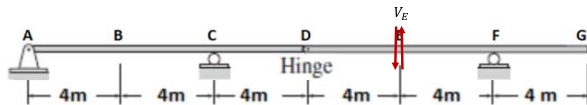


Remove the ability to resist Moment at point E.
Take note that there is no movement at the support.

Rotate wrt the specified direction of moment.



Influence Line for shear at E



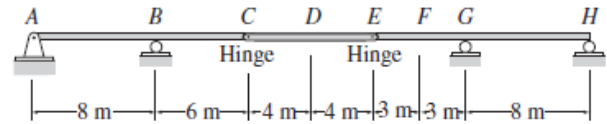
Remove the ability to resist shear at point B. Take note that there is no movement at the support.

Drag wrt the direction of force

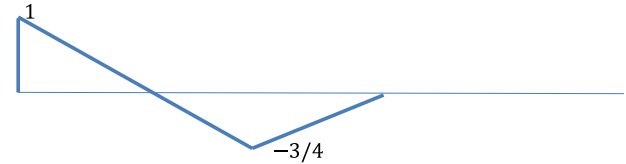


Problem#4: (P8.33-34, Structural analysis, Aslam Kassimali 4th ed.)

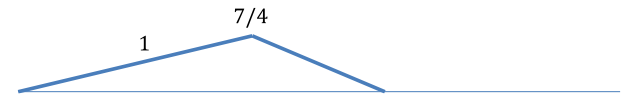
Draw the influence lines for the vertical reaction at the supports, and the shear and bending moments at point D beam shown.



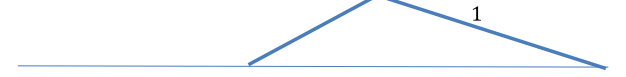
Influence for Reaction at A



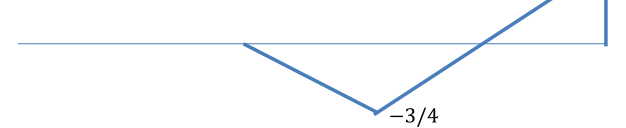
Influence Line for Reaction at C



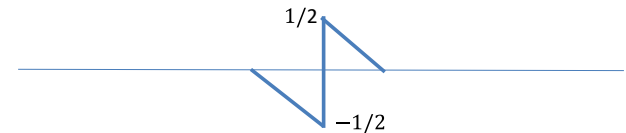
Influence Line for Reaction at G



Influence Line for Reaction at H



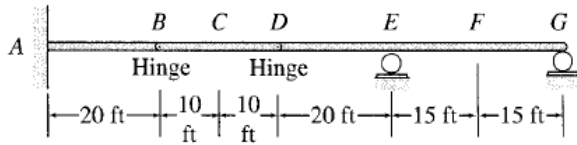
Influence Line for Shear at D



Influence Line for Moment at D

Problem#5: (P8.29-30, Structural analysis, Aslam Kassimali 4th ed.)

Draw the influence lines for the reaction moment at support A and the vertical reaction at supports A, E, and G. Shear and bending moment at points C of the beam shown.



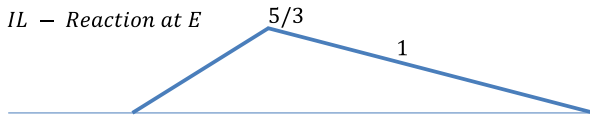
Influence Line for Reaction at A



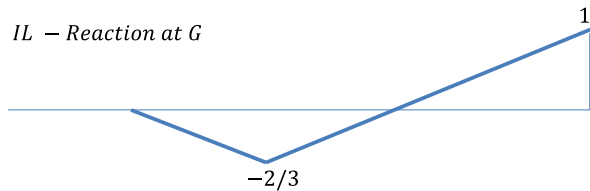
Influence Line for Moment at A



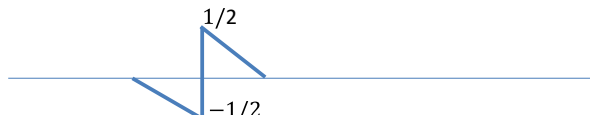
IL – Reaction at E



IL – Reaction at G



IL – Shear at C



IL – Moment at C

